

**1** The transformation  $R$  of the plane is reflection in the line  $x = 0$ .

**(a)** Write down the matrix  $\mathbf{M}$  associated with  $R$ . **[1]**

**(b)** Find  $\mathbf{M}^2$ . **[1]**

**(c)** Interpret the result of part **(b)** in terms of the transformation  $R$ . **[1]**